

Constant-Lagrangian inflow fits of the full SPARC sample:

overall classification of 175 rotation curves

Synthesis of the category A–E analyses, prepared for E.P.J. de Haas

Summary

This report combines the five category analyses (A–E) of the SPARC rotation curves (175 galaxies) into one physical classification. It is based only on the stored results of those analyses; nothing was refitted. Every galaxy is assigned to the simplest CL description that the respective analysis accepted, using the CL inflow model and its extensions from [1,2]. The result is:

- **Single-L** (100): one constant Lagrangian (M, R) describes the whole curve.
- **Double** (25): a Lagrangian reset (two-L, nested spiral) or the equivalent rising window.
- **Single-L + descent** (24): a CL curve followed by a Keplerian descent window.
- **Descent only** (7): the inner CL bulge is unresolved and the curve is a compact central mass plus a Keplerian descent.
- **Multi-region** (6): two windows, or a reset plus a window.
- **Flattening window** (2).
- **Unresolved** (11): bulge-dominated or non-nested galaxies that the present extensions describe only with unphysical parameters.

The classes form a clear morphological sequence (Kruskal–Wallis on Hubble type for the four main classes, $p = 6e - 12$). Single-L and double galaxies are late types (median Sm), descent galaxies are intermediate spirals (median Sbc), descent-only and unresolved galaxies are early types (median Sb and Sab). Over the full sample the median χ^2_ν goes from 0.86 (single-L) to 0.50 (final description) and the median RMS_{rel} in v^2 from 0.135 to 0.097. 19 galaxies remain at $\chi^2_\nu > 2$; apart from NGC 4051 (single-L, $n = 7$) all are high-precision curves, either massive bulged spirals with a sharp inner peak or well-sampled nearby disks that need more than one reset.

1 Basis of the synthesis

Models. All descriptions use the CL inflow quantities $X = \sqrt{2GM/R} - H_z R$ and $Y(r) = \sqrt{2GM/r} - H_z r$ with $H_z = 2.2 \times 10^{-18} \text{ s}^{-1}$, fitted to v^2 with $\sigma_{v^2} = 2V_{\text{obs}}\delta V_{\text{obs}}$ (SPARC [3]).

- **Single-L** [2, Eqs. 17–18]: $v^2 = \frac{1}{2}X^2r^2/R^2$ for $r \leq R$ and $\frac{3}{2}X^2 - Y^2$ beyond; asymptotic speed $v_L = \sqrt{3/2}X$.
- **Two-L** [2, Eqs. 24–26]: $L_1(M_1, R_1)$ up to R_2 , then $L_2(M_2, R_2)$.
- **Virial window** [1, Eq. 6; 2, Eq. 19]: $+p[\frac{1}{2}Y^2 - \Phi_p]$ beyond r_v . With the continuous gauge and $H_z \rightarrow 0$ this is $v^2 = A + GM_N/r$, with $M_N = (p - 2)M$. So $p > 2$ is a Keplerian descent, $0 < p < 2$ a flattening and $p < 0$ an additional rise.
- **Two windows** [1, Eqs. 5–7].

Input analyses. **A** (54): single-L adequate, with grades I–III. **B** (45): $+-+$ wave, two-L tested; accepted at $\Delta\text{BIC} < -6$, with physical screening (nested / needs more regions / not nested). **C** (41): $-+-$ wave, virial window tested against single-L and two-L. **D** (18): other structure, five models tested. **E** (17): poor fit without detected structure (low-power runs test), extensions tested directly by BIC.

Assignment rules. Each galaxy keeps the outcome of its own analysis, mapped as follows.

Descent galaxies from C are labelled *descent only* when no data point lies inside the fitted CL bulge radius and $p \geq 30$ (inner Lagrangian mass negligible against M_N), or when at most two points precede r_v . NGC 7814 (D) is assigned to this class because its descent starts at the first data points.

Table 1: Mapping of the category outcomes to the overall classes, and counts.

Overall class	Sub-type	A	B	C	D	E	Total
Single-L	adequate	54					54
	extra scatter (σ_{int})					6	6
	structure within errors				6		6
	undecided ($-6 \leq \text{dBIC} < -2$)		7				7
	weak + - + wave		15				15
	weak - + - wave			12			12
Double (two-L reset)	nested two-L		13				13
	rising window ($p < 0$)			1	1	5	7
	two-L, needs more regions		5				5
Single-L + descent	CL + Kepler descent			20	1	3	24
Descent only	descent from first points				1		1
	descent-dominated			6			6
Multi-region	reset + disk window				2		2
	two windows				2	2	4
Flattening window	$0 < p < 2$			2			2
Unresolved (bulge / non-nested)	extreme window parameters				5	1	6
	two-L non-nested		5				5
Total		54	45	41	18	17	175

2 The classes

2.1 Single-L (100)

n median 10; χ^2_ν 0.28→0.28; RMS_{rel} 0.109→0.109; $Q=1$: 47/100; final $\chi^2_\nu > 2$: 1; $n \leq 13$ (tentative): 69; asymptotic speed/ $V_{\text{flat}} = 1.08$ ($r=0.99$, 58 galaxies).

One constant Lagrangian describes the full rotation curve. The class consists of the 54 category-A galaxies plus 46 galaxies from B–E whose residual structure, although detectable in sign, does not justify extra parameters ($\Delta\text{BIC} \geq -6$). These comprise 7 B galaxies with positive but not strong evidence for a reset (NGC 3109, NGC 3972, NGC 4010, UGC 6446, UGC 7125, UGC 7323, UGC 7399), weak waves in B and C, structure within the errors in D, and E galaxies whose excess χ^2 corresponds to an intrinsic scatter of a few km s^{-1} . The class is dominated by late types and dwarfs (median Sm; 55 Sm/Im/BCD). The curves are short: 69 of 100 have $n \leq 13$, so “single-L adequate” often means that the data cannot demand more. The best-determined quantity is v_L , which tracks V_{flat} closely. M and R are strongly correlated (category-A median $\rho_{MR} = 0.97$), and only 17 category-A galaxies have both determined to better than 30% (grade I).

2.2 Double: Lagrangian reset (25)

n median 21; χ^2_ν 3.26→0.90; RMS_{rel} 0.249→0.127; $Q=1$: 16/25; final $\chi^2_\nu > 2$: 6; $n \leq 13$ (tentative): 8; asymptotic speed/ $V_{\text{flat}} = 1.14$ ($r=0.95$, 16 galaxies).

The rotation curve switches from an inner Lagrangian (bulge/bar zone, L_1) to an outer one (disk, L_2) at R_2 , the nested-spiral configuration. The class contains the 13 clean nested two-L galaxies of B, the 5 B galaxies improved by two-L but needing further regions (DDO154, NGC1003, NGC2403, NGC5585, UGC03580), and 7 galaxies from C, D and E where a rising window ($p < 0$) is preferred; in most of these the two-L fit is statistically equivalent. It is the late-type class (median Sm, 20 of 25 Sd–Im; photometric bulge in 2). Typical scales: $R_1/R_2 \approx 0.3$, $M_2/M_1 \approx 8$, $v_{L2}/v_{L1} \approx 1.6$, R_2 at 1–4 disk scale lengths. Caveat: R_2 is a discontinuity location and is localised only to the gap between data points; the size of the v^2 jump is not constrained by the data.

2.3 Single-L + descent (24)

n median 29; χ^2_ν 4.28→0.89; RMS_{rel} 0.149→0.075; $Q=1$: 18/24; final $\chi^2_\nu > 2$: 5; $n \leq 13$ (tentative): 6; asymptotic speed/ $V_{\text{flat}} = 0.94$ ($r=0.96$, 20 galaxies).

A CL rise and peak followed, beyond r_v , by a Keplerian descent $v^2 = A + GM_N/r$ onto a floor A . These are intermediate spirals (median Sbc; photometric bulge in 10 of 24). The descent starts at $r_v \approx 3$ –4 disk scale lengths, just beyond the peak. The floor and window radius are well determined (category C: 6% and 9%), the descent mass less so (23%), and the descent mass is of order of the baryonic mass. The virial strength p ranges

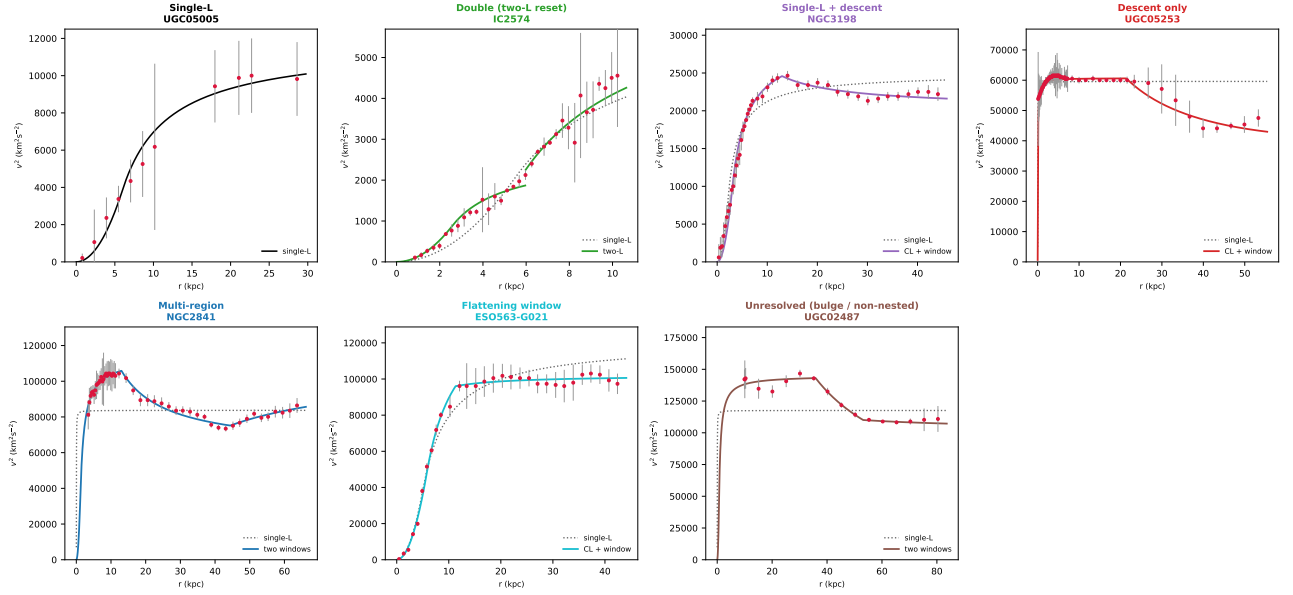


Figure 1: One representative per class, drawn from the stored best fits (dotted: single-L; solid: final description).

widely (category C: $p \approx 3.8\text{--}60$, 16–84%); the pure Kepler case $p = 3$ is one end of this range, not the rule. Residual structure remains in 12 galaxies, located at the inner peak of high-precision curves.

2.4 Descent only (7)

n median 19; χ^2_ν 5.54→1.06; RMS_{rel} 0.114→0.053; $Q=1$: 5/7; final $\chi^2_\nu > 2$: 2; $n \leq 13$ (tentative): 0; asymptotic speed/ $V_{\text{flat}} = 0.80$ ($r=0.98$, 6 galaxies).

NGC0891, NGC5033, NGC5371, NGC6674, UGC02916, UGC05253, NGC7814. All are Sab–Sc spirals (Sb, Sc, Sbc, Sb, Sab, Sab, Sab), with a photometric bulge in 6 of 7. The inner CL Lagrangian is not resolved: its radius lies inside the first data point, or the descent begins at the first points. The curve is described by a compact central mass with a Keplerian descent onto a floor. The quantities r_v , A and M_N are well defined; M , R and p are not. The floor lies well below V_{flat} (median ratio 0.80) because these curves are still descending at the last point. This class is the direct successor of the earlier descent-only fits; with the window formulation the parameters are positive and remain defined for $H_z \neq 0$.

2.5 Multi-region (6)

n median 23; χ^2_ν 4.94→1.14; RMS_{rel} 0.133→0.052; $Q=1$: 4/6; final $\chi^2_\nu > 2$: 1; $n \leq 13$ (tentative): 1.

NGC1090, NGC2841, NGC6015, UGC00128, NGC6195, UGC05764. These need two windows or a reset plus a window. NGC 2841 (descent at 13 kpc, renewed rise at 44 kpc) and NGC 6015 (reset at 2.3 kpc, flattening at 3.5 kpc) are clean examples of the multi-region configurations of [1]. The others are improved but weakly constrained.

2.6 Flattening window (2)

n median 31; χ^2_ν 2.34→0.76; RMS_{rel} 0.099→0.070; $Q=1$: 2/2; final $\chi^2_\nu > 2$: 0; $n \leq 13$ (tentative): 0.

ESO563-G021 and IC 4202 (Sbc, edge-on): beyond $r_v \approx 10$ kpc the curve rises more slowly than single-L ($0 < p < 2$), with no descent.

2.7 Unresolved (11)

n median 28; χ^2_ν 3.06→0.80; RMS_{rel} 0.125→0.058; $Q=1$: 7/11; final $\chi^2_\nu > 2$: 4; $n \leq 13$ (tentative): 2.

NGC3726, NGC4013, UGC02885, UGC06614, UGC06787, NGC0289, UGC02487, UGC03546, UGC06786, UGC09133, NGC2955. All are early or intermediate types (S0–Sc, median Sab), with a photometric bulge in 9 of 11. Statistically the extensions improve them strongly, but only with non-nested resets (a v^2 drop at R_2 , or $M_2 < M_1$) or extreme window strengths with the CL bulge radius at its lower bound. Their residuals trace a sharp inner peak that the parabolic CL bulge ($v^2 \propto r^2$ inside R) cannot follow. This class marks the present limit of the model, not a separate dynamical regime.

3 Morphological sequence

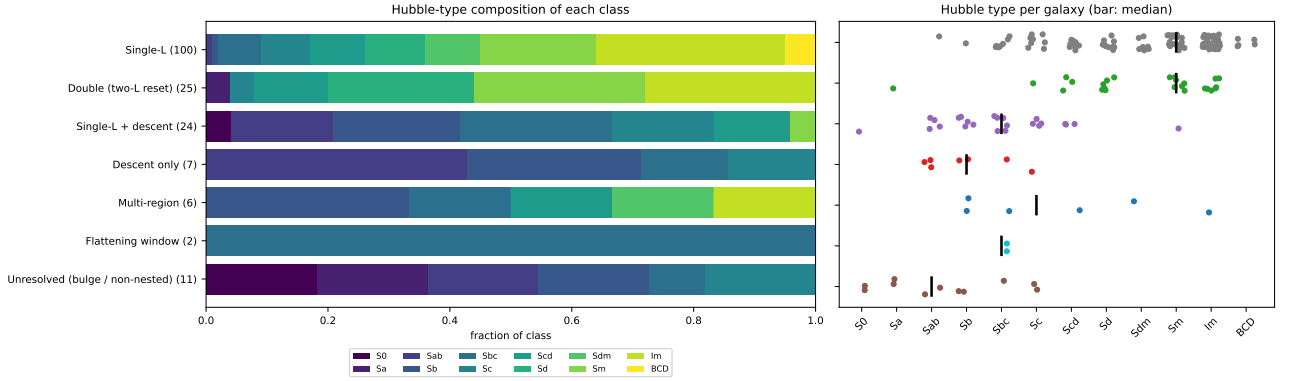


Figure 2: Hubble-type composition of the classes (left) and Hubble type per galaxy (right; bar = median).

The classes are ordered in morphology (Fig. 2). Single-L and double galaxies are late types and dwarfs (medians Sm and Sm). Single-L + descent galaxies are intermediate spirals (median Sbc). Descent-only and unresolved galaxies are the earliest types (medians Sb and Sab). The difference between the four main classes is highly significant (Kruskal–Wallis $H = 55.3$, $p = 6e - 12$). The fraction of galaxies with a photometric bulge rises along the same sequence: 2

4 Global fit quality

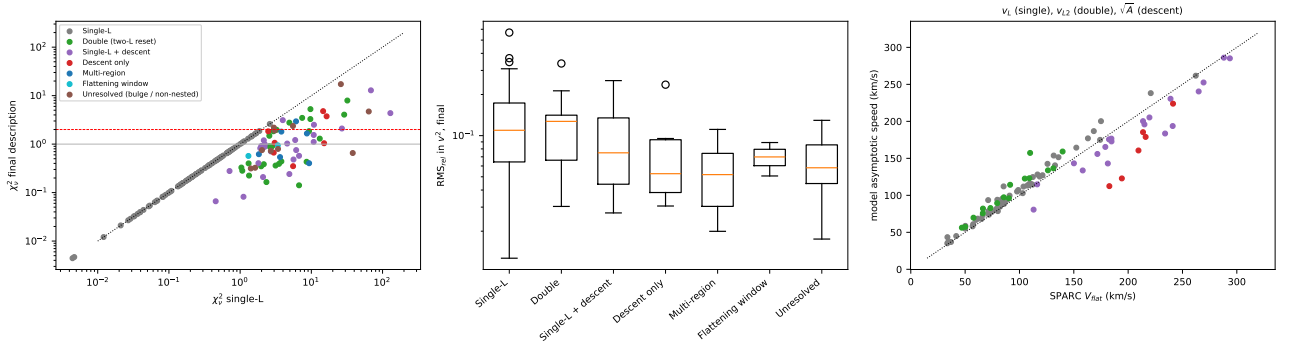


Figure 3: Left: χ^2_{ν} of single-L versus the final description. Middle: final RMS_{rel} per class. Right: model asymptotic speed (v_L for single-L, v_{L2} for nested two-L, $\sqrt{A}$ for descents) versus SPARC V_{flat} .

Table 2: Fit quality per class (medians).

Class	N	n	χ^2_{ν} single-L	χ^2_{ν} final	RMS_{rel} final	final $\chi^2_{\nu} > 2$	$n \leq 13$
Single-L	100	10	0.28	0.28	0.109	1	69
Double (two-L reset)	25	21	3.26	0.90	0.127	6	8
Single-L + descent	24	29	4.28	0.89	0.075	5	6
Descent only	7	19	5.54	1.06	0.053	2	0
Multi-region	6	23	4.94	1.14	0.052	1	1
Flattening window	2	31	2.34	0.76	0.070	0	0
Unresolved (bulge / non-nested)	11	28	3.06	0.80	0.058	4	2
All	175	14	0.86	0.50	0.097	19	86

With the final descriptions, 156 of 175 galaxies have $\chi^2_{\nu} \leq 2$, and the median RMS_{rel} in v^2 is 0.097. The 19 galaxies above $\chi^2_{\nu} = 2$ are DDO154, NGC1003, NGC2403, NGC3726, NGC4013, NGC5585, UGC03580, UGC06787, NGC2903, NGC5055, NGC5371, NGC5907, NGC6674, NGC6946, UGC02953, UGC11455, UGC00128, UGC09133, NGC4051. Apart from NGC 4051 ($n = 7$, single-L with one deviant point), they are high-precision curves of two kinds: massive bulged spirals with a sharp inner peak (e.g. NGC 5055, UGC 2953, UGC 6787), and well-sampled nearby disks whose curves need more than one reset (NGC 2403, NGC 5585, NGC 1003,

DDO 154). SPARC errors are conservative on average (median $\chi^2_\nu < 1$ in all well-fitting classes), so χ^2_ν is a weak discriminator within classes; RMS_{rel} , residual structure and ΔBIC are the more informative measures. The model asymptotic speed tracks V_{flat} in every class (right panel). The systematic offsets reflect whether the curve is still rising (single-L, double: model asymptote above V_{flat}) or still descending (descent classes: floor below V_{flat}) at the last measured point.

5 Limitations

1. **Thresholds.** Class boundaries depend on $\Delta\text{BIC} < -6$ and on the physical screening rules. The seven undecided B galaxies and the borderline rising-window galaxies (NGC 300, NGC 6946) could move between Single-L and Double with more data.
2. **Short curves.** 86 galaxies have $n \leq 13$. For these the sign-based screen has little power, and a “single-L” classification often reflects data limitations.
3. **Heterogeneous selection.** The five analyses tested slightly different model sets: B tested two-L only, while C–E compared windows and two-L. A uniform re-run with all models on all galaxies would remove this dependence; by design it was not done here.
4. **Bulge description.** The unresolved class and most remaining residual structure point to the inner CL bulge. An explicit bulge component is the main open refinement.
5. **Model features.** The v^2 jump at a Lagrangian reset is not constrained by data. Window strengths p are degenerate with M when the descent dominates, so M_N and A should be quoted instead.
6. **Interpretation.** Associating windows with galaxy history, as proposed in [1], requires independent indicators (morphology, kinematic asymmetry, HI warps, environment); these were not part of the analyses.

6 Conclusions

Starting from a two-parameter constant Lagrangian, 100 of 175 SPARC galaxies need nothing more. A further 56 are described by one additional, physically interpretable feature: a Lagrangian reset (late types) or a Keplerian descent (bulged spirals). 8 need two features or a flattening, and 11 bulge-dominated galaxies remain beyond the present model. The ordering of these classes along the Hubble sequence is the main empirical result. It ties the CL description to galaxy structure: the number and kind of Lagrangian regions follow the prominence of the bulge.

References

- [1] E.P.J. de Haas (2026), From Rotation Curves to Cosmic Time: Probing High-Redshift Expansion through Nested Spiral Dynamics, *J. High Energy Phys. Grav. Cosmol.* 12, 334–367, doi:10.4236/jhepgc.2026.121022.
- [2] E.P.J. de Haas, Galactic Rotation Curves and the Constant–Lagrangian Field: Empirical Tests within the Q_g Rotor Framework (manuscript; single-L Eqs. 17–18, virial term Eq. 19, two-L Eqs. 24–26).
- [3] F. Lelli, S.S. McGaugh, J.M. Schombert (2016), SPARC, *AJ* 152, 157.
Category reports A–E (this project): CL_categoryA_report, CL_categoryB_report, CL_categoryC_report, CL_categoryD_report, CL_categoryE_report.

Table 3: All 175 SPARC galaxies: overall class, origin, sub-type and fit quality (single-L $\rightarrow$ final). t = tentative ($n \leq 13$).

Galaxy	Class	Sub-type	Cat.	Type	Q	n	χ^2_L	S	χ^2_L	final	RMS final	Δ BIC	Quality / grade
CamB t	Single-L	adequate	A	Im	2	9	0.04	0.04	0.04	0.04	0.044	0.0	grade III
D512-2 t	Single-L	adequate	A	Im	2	4	0.07	0.07	0.07	0.07	0.055	0.0	grade II
D564-8 t	Single-L	adequate	A	Im	2	6	0.15	0.15	0.15	0.15	0.092	0.0	grade II
DDO170 t	Single-L	adequate	A	Im	2	8	0.50	0.50	0.50	0.50	0.076	0.0	grade II
F561-1 t	Single-L	adequate	A	Sm	3	6	0.01	0.01	0.01	0.01	0.021	0.0	grade II
F563-V1 t	Single-L	adequate	A	Im	3	6	0.14	0.14	0.14	0.14	0.110	0.0	grade III
F563-V2 t	Single-L	adequate	A	Im	1	10	0.03	0.03	0.03	0.03	0.048	0.0	grade I
F565-V2 t	Single-L	adequate	A	Im	2	7	0.04	0.04	0.04	0.04	0.085	0.0	grade II
F567-2 t	Single-L	adequate	A	Sm	3	5	0.15	0.15	0.15	0.15	0.089	0.0	grade II
F571-V1 t	Single-L	adequate	A	Sd	2	7	0.04	0.04	0.04	0.04	0.060	0.0	grade II
F574-2 t	Single-L	adequate	A	Sm	3	5	0.03	0.03	0.03	0.03	0.143	0.0	grade III
F583-1	Single-L	adequate	A	Sm	1	25	0.21	0.21	0.21	0.21	0.250	0.0	grade I
F583-4 t	Single-L	adequate	A	Sc	1	12	0.56	0.56	0.56	0.56	0.239	0.0	grade I
NGC3877 t	Single-L	adequate	A	Sc	2	13	0.15	0.15	0.15	0.15	0.062	0.0	grade I
NGC3917	Single-L	adequate	A	Scd	1	17	0.40	0.40	0.40	0.40	0.040	0.0	grade I
NGC3949 t	Single-L	adequate	A	Sbc	2	7	0.03	0.03	0.03	0.03	0.019	0.0	grade II
NGC3953 t	Single-L	adequate	A	Sbc	1	8	0.52	0.52	0.52	0.52	0.048	0.0	grade II
NGC4068 t	Single-L	adequate	A	Im	2	6	0.08	0.08	0.08	0.08	0.077	0.0	grade III
NGC4085 t	Single-L	adequate	A	Sc	2	7	0.13	0.13	0.13	0.13	0.038	0.0	grade II
NGC4389 t	Single-L	adequate	A	Sbc	3	6	0.75	0.75	0.75	0.75	0.176	0.0	grade II
NGC6789 t	Single-L	adequate	A	BCD	2	4	0.58	0.58	0.58	0.58	0.258	0.0	grade III
UGC02023 t	Single-L	adequate	A	Im	2	5	0.08	0.08	0.08	0.08	0.172	0.0	grade III
UGC02259 t	Single-L	adequate	A	Sdm	2	8	0.66	0.66	0.66	0.66	0.052	0.0	grade II
UGC04325 t	Single-L	adequate	A	Sm	1	8	0.41	0.41	0.41	0.41	0.049	0.0	grade II
UGC04483 t	Single-L	adequate	A	Im	2	8	0.17	0.17	0.17	0.17	0.188	0.0	grade I
UGC04499 t	Single-L	adequate	A	Sdm	1	9	0.35	0.35	0.35	0.35	0.100	0.0	grade I
UGC05005 t	Single-L	adequate	A	Im	1	11	0.19	0.19	0.19	0.19	0.278	0.0	grade I
UGC05414 t	Single-L	adequate	A	Im	1	6	0.93	0.93	0.93	0.93	0.137	0.0	grade II
UGC05750 t	Single-L	adequate	A	Sdm	1	11	0.04	0.04	0.04	0.04	0.065	0.0	grade I
UGC05918 t	Single-L	adequate	A	Im	2	8	0.19	0.19	0.19	0.19	0.129	0.0	grade II
UGC05999 t	Single-L	adequate	A	Im	2	5	0.02	0.02	0.02	0.02	0.013	0.0	grade II
UGC06399 t	Single-L	adequate	A	Sm	1	9	0.22	0.22	0.22	0.22	0.111	0.0	grade I
UGC06628 t	Single-L	adequate	A	Sm	2	7	0.00	0.00	0.00	0.00	0.025	0.0	grade II
UGC06667 t	Single-L	adequate	A	Scd	1	9	0.72	0.72	0.72	0.72	0.145	0.0	grade I
UGC06818 t	Single-L	adequate	A	Sm	2	8	0.68	0.68	0.68	0.68	0.274	0.0	grade I
UGC06917 t	Single-L	adequate	A	Sm	1	11	0.74	0.74	0.74	0.74	0.086	0.0	grade II
UGC06923 t	Single-L	adequate	A	Im	2	6	0.71	0.71	0.71	0.71	0.226	0.0	grade II
UGC06930 t	Single-L	adequate	A	Sd	1	10	0.10	0.10	0.10	0.10	0.044	0.0	grade II
UGC06973 t	Single-L	adequate	A	Sab	3	9	0.36	0.36	0.36	0.36	0.036	0.0	grade II
UGC07089 t	Single-L	adequate	A	Sdm	2	12	0.70	0.70	0.70	0.70	0.199	0.0	grade I
UGC07151 t	Single-L	adequate	A	Scd	1	11	0.45	0.45	0.45	0.45	0.070	0.0	grade I
UGC07232 t	Single-L	adequate	A	Im	2	4	0.71	0.71	0.71	0.71	0.186	0.0	grade III
UGC07261 t	Single-L	adequate	A	Sdm	2	7	0.09	0.09	0.09	0.09	0.041	0.0	grade II
UGC07559 t	Single-L	adequate	A	Im	2	7	0.11	0.11	0.11	0.11	0.152	0.0	grade II
UGC07577 t	Single-L	adequate	A	Im	2	9	0.06	0.06	0.06	0.06	0.272	0.0	grade III
UGC07603 t	Single-L	adequate	A	Sd	1	12	0.46	0.46	0.46	0.46	0.126	0.0	grade I
UGC07608 t	Single-L	adequate	A	Im	1	8	0.16	0.16	0.16	0.16	0.092	0.0	grade II
UGC07690 t	Single-L	adequate	A	Im	2	7	0.66	0.66	0.66	0.66	0.103	0.0	grade II
UGC07866 t	Single-L	adequate	A	Im	2	7	0.21	0.21	0.21	0.21	0.170	0.0	grade II
UGC08837 t	Single-L	adequate	A	Im	2	8	0.05	0.05	0.05	0.05	0.041	0.0	grade I
UGC09992 t	Single-L	adequate	A	Im	2	5	0.00	0.00	0.00	0.00	0.018	0.0	grade III
UGC10310 t	Single-L	adequate	A	Sm	1	7	0.07	0.07	0.07	0.07	0.049	0.0	grade II
UGC11557 t	Single-L	adequate	A	Sdm	2	12	0.18	0.18	0.18	0.18	0.308	0.0	grade I
UGCA281 t	Single-L	adequate	A	BCD	3	7	0.29	0.29	0.29	0.29	0.186	0.0	grade II
NGC3893 t	Single-L	extra scatter (σ_{int})	E	Sc	1	10	1.45	1.45	1.45	1.45	0.128	0.0	single-L kept

Galaxy	Class	Sub-type	Cat.	Type	Q	n	χ^2_S	χ^2_{ν} final	RMS final	ΔBIC	Quality / grade
NGC4051 t	Single-L	extra scatter (σ_{int})	E	Sbc	2	7	2.61	2.61	0.270	0.0	single-L kept
UGC00634 t	Single-L	extra scatter (σ_{int})	E	Sm	2	4	1.86	1.86	0.035	0.0	single-L kept
UGC00891 t	Single-L	extra scatter (σ_{int})	E	Sm	2	5	1.07	1.07	0.108	0.0	single-L kept
UGC02455 t	Single-L	extra scatter (σ_{int})	E	Im	3	8	1.60	1.60	0.369	0.0	single-L kept
UGCA442 t	Single-L	extra scatter (σ_{int})	E	Sm	1	8	1.47	1.47	0.214	0.0	single-L kept
DDO064	Single-L	structure within errors	D	Im	1	14	0.24	0.24	0.306	0.0	single-L adequate
F563-1	Single-L	structure within errors	D	Sm	1	17	0.24	0.24	0.166	0.0	single-L adequate
NGC0024	Single-L	structure within errors	D	Sc	1	29	0.24	0.24	0.085	0.0	single-L adequate
NGC2915	Single-L	structure within errors	D	BCD	2	30	0.26	0.26	0.123	0.0	single-L adequate
PGC51017 t	Single-L	structure within errors	D	BCD	3	6	0.21	0.21	0.096	0.0	single-L adequate
UGC01281	Single-L	structure within errors	D	Sdm	1	25	0.03	0.03	0.065	0.0	single-L adequate
NGC3109	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sm	1	25	0.78	0.78	0.127	0.0	structured residuals
NGC3972 t	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sbc	1	10	1.42	1.42	0.138	0.0	structured residuals
NGC4010 t	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sd	2	12	1.69	1.69	0.239	0.0	structured residuals
UGC06446	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sd	1	17	0.67	0.67	0.147	0.0	structured residuals
UGC07125 t	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sm	1	13	1.15	1.15	0.150	0.0	structured residuals
UGC07323 t	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sdm	1	10	1.68	1.68	0.284	0.0	structured residuals
UGC07399 t	Single-L	undecided ($-6 \leq \text{dBIC} < -2$)	B	Sdm	1	10	1.29	1.29	0.098	0.0	structured residuals
ESO079-G014	Single-L	weak + + + wave	B	Sbc	1	15	1.12	1.12	0.345	0.0	structured residuals
F568-1 t	Single-L	weak + + + wave	B	Sc	1	12	0.05	0.05	0.094	0.0	structured residuals
F568-3	Single-L	weak + + + wave	B	Sd	1	18	0.19	0.19	0.092	0.0	structured residuals
F574-1	Single-L	weak + + + wave	B	Sd	1	14	0.16	0.16	0.109	0.0	structured residuals
F579-V1	Single-L	weak + + + wave	B	Sc	1	14	0.10	0.10	0.108	0.0	structured residuals
KK98-251	Single-L	weak + + + wave	B	Im	2	15	0.18	0.18	0.066	0.0	structured residuals
NGC0100	Single-L	weak + + + wave	B	Scd	1	21	0.37	0.37	0.146	0.0	structured residuals
NGC2976	Single-L	weak + + + wave	B	Sc	2	27	0.47	0.47	0.155	0.0	structured residuals
NGC4214	Single-L	weak + + + wave	B	Im	2	14	0.53	0.53	0.122	0.0	structured residuals
NGC5005	Single-L	weak + + + wave	B	Sbc	1	18	0.20	0.20	0.087	0.0	structured residuals
UGC05829 t	Single-L	weak + + + wave	B	Im	2	11	0.83	0.83	0.286	0.0	structured residuals
UGC08286	Single-L	weak + + + wave	B	Scd	1	17	0.15	0.15	0.028	0.0	structured residuals
UGC08550 t	Single-L	weak + + + wave	B	Sd	1	11	0.68	0.68	0.061	0.0	structured residuals
UGC12632	Single-L	weak + + + wave	B	Sm	1	15	0.28	0.28	0.179	0.0	structured residuals
UGCA444	Single-L	weak + + + wave	B	Im	2	36	0.26	0.26	0.223	0.0	structured residuals
F568-V1	Single-L	weak - + - wave	C	Sd	1	15	0.08	0.08	0.145	0.0	structured residuals
NGC1705	Single-L	weak - + - wave	C	BCD	3	14	0.15	0.15	0.058	0.0	structured residuals
NGC2366	Single-L	weak - + - wave	C	Im	3	26	0.47	0.47	0.165	0.0	structured residuals
NGC3769 t	Single-L	weak - + - wave	C	Sb	2	12	0.79	0.79	0.097	0.0	structured residuals
NGC4183	Single-L	weak - + - wave	C	Scd	1	23	0.54	0.54	0.145	0.0	structured residuals
NGC4559	Single-L	weak - + - wave	C	Scd	1	32	0.44	0.44	0.155	0.0	structured residuals
UGC01230 t	Single-L	weak - + - wave	C	Sm	1	11	0.37	0.37	0.128	0.0	structured residuals
UGC04305	Single-L	weak - + - wave	C	Im	3	22	0.65	0.65	0.568	0.0	structured residuals
UGC05721	Single-L	weak - + - wave	C	Sd	1	23	1.01	1.01	0.222	0.0	structured residuals
UGC06983	Single-L	weak - + - wave	C	Scd	1	17	0.32	0.32	0.056	0.0	structured residuals
UGC08490	Single-L	weak - + - wave	C	Sm	1	30	0.50	0.50	0.103	0.0	structured residuals
UGC09037	Single-L	weak - + - wave	C	Scd	2	22	0.86	0.86	0.091	0.0	structured residuals
D631-7	Double (two-L reset)	nested two-L	B	Im	1	16	1.33	0.22	0.049	-10.4	grade I
DDO161	Double (two-L reset)	nested two-L	B	Im	1	31	2.15	0.38	0.096	-45.1	grade I
ESO116-G012	Double (two-L reset)	nested two-L	B	Sd	1	15	2.57	1.49	0.194	-11.7	grade II
F571-8 t	Double (two-L reset)	nested two-L	B	Sc	1	13	3.79	0.44	0.132	-32.6	grade II
IC2574	Double (two-L reset)	nested two-L	B	Sm	2	34	13.13	1.30	0.107	-374.1	grade I
NGC0055	Double (two-L reset)	nested two-L	B	Sm	2	21	1.04	0.33	0.056	-8.1	grade II
NGC0247	Double (two-L reset)	nested two-L	B	Sd	2	26	8.58	0.43	0.038	-189.9	grade II
NGC3741	Double (two-L reset)	nested two-L	B	Im	1	21	3.52	0.39	0.134	-54.0	grade I
NGC7793	Double (two-L reset)	nested two-L	B	Sd	1	46	2.57	0.90	0.213	-67.5	grade I
UGC00731 t	Double (two-L reset)	nested two-L	B	Im	1	12	2.34	0.16	0.030	-17.1	grade II
UGC04278	Double (two-L reset)	nested two-L	B	Sd	1	25	1.97	0.35	0.141	-31.6	grade I
UGC07524	Double (two-L reset)	nested two-L	B	Sm	1	31	1.07	0.28	0.127	-16.6	grade II

Galaxy	Class	Sub-type	Cat.	Type	Q	n	χ^2_S	χ^2_{ν}	RMS final	ΔBIC	Quality / grade
UGC12732	Double (two-L reset)	nested two-L	B	Sm	1	16	2.81	0.90	0.127	-23.0	grade II
DDO168 t	Double (two-L reset)	rising window (p<0)	E	Im	2	10	2.92	1.84	0.142	-7.7	weakly constrained
ESO444-G084 t	Double (two-L reset)	rising window (p<0)	E	Im	2	7	3.14	0.36	0.066	-10.7	weakly constrained
NGC0300	Double (two-L reset)	rising window (p<0)	D	Sd	2	25	1.31	0.40	0.065	-15.4	clean
NGC6946	Double (two-L reset)	rising window (p<0)	C	Scd	1	58	9.71	5.27	0.173	-251.4	rising window
UGC00191 t	Double (two-L reset)	rising window (p<0)	E	Sm	1	9	3.26	1.97	0.109	-8.5	weakly constrained
UGC05716 t	Double (two-L reset)	rising window (p<0)	E	Sm	2	12	9.28	1.85	0.049	-73.0	clean
UGC11820 t	Double (two-L reset)	rising window (p<0)	E	Sm	1	10	6.72	0.14	0.159	-48.3	clean
DDO154 t	Double (two-L reset)	two-L, needs more regions	B	Im	2	12	7.43	3.48	0.141	-41.6	grade IIM
NGC1003	Double (two-L reset)	two-L, needs more regions	B	Scd	1	36	4.91	2.77	0.114	-71.2	grade IM
NGC2403	Double (two-L reset)	two-L, needs more regions	B	Scd	1	73	32.04	7.93	0.105	-1719.3	grade IM
NGC5585	Double (two-L reset)	two-L, needs more regions	B	Sd	1	24	28.93	4.05	0.127	-549.1	grade IIM
UGC03580	Double (two-L reset)	two-L, needs more regions	B	Sa	2	47	9.69	3.29	0.337	-286.9	grade IM
NGC0801 t	Single-L + descent	CL + Kepler descent	E	Sc	1	13	10.80	1.54	0.074	-99.8	clean
NGC2683 t	Single-L + descent	CL + Kepler descent	E	Sb	2	11	10.79	1.11	0.088	-84.6	weakly constrained
NGC2903	Single-L + descent	CL + Kepler descent	C	Sbc	1	34	26.86	2.10	0.100	-789.5	grade IM
NGC2998 t	Single-L + descent	CL + Kepler descent	E	Sc	1	13	1.98	0.91	0.253	-8.5	weakly constrained
NGC3198	Single-L + descent	CL + Kepler descent	C	Sc	1	43	6.13	0.75	0.224	-214.2	grade I
NGC3521	Single-L + descent	CL + Kepler descent	C	Sbc	1	41	0.45	0.07	0.039	-7.8	grade III
NGC3992 t	Single-L + descent	CL + Kepler descent	C	Sbc	1	9	5.62	0.48	0.029	-32.5	grade I
NGC4088 t	Single-L + descent	CL + Kepler descent	C	Sbc	1	12	1.81	0.40	0.076	-9.9	grade III
NGC4100	Single-L + descent	CL + Kepler descent	C	Sbc	1	24	6.72	0.57	0.050	-130.0	grade I
NGC4138 t	Single-L + descent	CL + Kepler descent	D	S0	2	7	4.91	0.24	0.072	-19.9	weakly constrained
NGC4157	Single-L + descent	CL + Kepler descent	C	Sb	1	17	2.10	0.21	0.052	-23.1	grade II
NGC4217	Single-L + descent	CL + Kepler descent	C	Sb	1	19	1.11	0.08	0.036	-11.8	grade III
NGC5055	Single-L + descent	CL + Kepler descent	C	Sbc	1	28	128.68	4.36	0.120	-3234.6	grade IM
NGC5907	Single-L + descent	CL + Kepler descent	C	Sc	1	19	10.88	2.50	0.036	-141.6	grade IIM
NGC5985	Single-L + descent	CL + Kepler descent	C	Sb	1	33	2.20	0.88	0.038	-35.9	grade III
NGC6503	Single-L + descent	CL + Kepler descent	C	Scd	1	31	1.92	0.82	0.046	-26.5	grade II
NGC7331	Single-L + descent	CL + Kepler descent	C	Sb	1	36	2.70	0.71	0.027	-62.2	grade II
UGC02953	Single-L + descent	CL + Kepler descent	C	Sab	2	115	68.64	12.87	0.248	-6317.8	grade IM
UGC03205	Single-L + descent	CL + Kepler descent	C	Sab	1	48	5.93	1.21	0.169	-211.8	grade II
UGC05986	Single-L + descent	CL + Kepler descent	C	Sm	2	15	2.15	1.20	0.126	-9.4	grade III
UGC08699	Single-L + descent	CL + Kepler descent	C	Sab	2	41	4.58	1.03	0.070	-133.3	grade II
UGC11455	Single-L + descent	CL + Kepler descent	C	Scd	1	36	3.97	3.12	0.170	-28.0	grade IIM
UGC11914	Single-L + descent	CL + Kepler descent	C	Sab	1	65	2.22	0.99	0.097	-70.9	grade III
UGC12506	Single-L + descent	CL + Kepler descent	C	Scd	2	31	0.71	0.28	0.163	-6.2	grade III
NGC7814	Descent only	descent from first points	D	Sab	1	18	2.96	0.67	0.031	-32.3	weakly constrained
NGC0891	Descent only	descent-dominated	C	Sb	1	18	3.06	1.06	0.091	-28.2	grade III
NGC5033	Descent only	descent-dominated	C	Sc	1	22	15.11	1.04	0.042	-277.3	grade III
NGC5371	Descent only	descent-dominated	C	Sbc	1	19	14.64	4.79	0.095	-171.2	grade IIM
NGC6674	Descent only	descent-dominated	C	Sb	1	15	16.38	3.74	0.236	-166.4	grade IIM
UGC02916	Descent only	descent-dominated	C	Sab	2	43	2.48	1.84	0.053	-22.5	grade III
UGC05253	Descent only	descent-dominated	C	Sab	2	73	5.54	0.35	0.034	-360.7	grade I
NGC1090	Multi-region	reset + disk window	D	Sbc	1	24	3.63	0.54	0.077	-57.5	weakly constrained
NGC6015	Multi-region	reset + disk window	D	Scd	2	44	8.66	1.66	0.064	-285.8	clean
NGC2841	Multi-region	two windows	D	Sb	1	50	9.31	0.40	0.020	-413.3	clean
NGC6195	Multi-region	two windows	E	Sb	1	23	1.83	0.62	0.039	-15.4	weakly constrained
UGC00128	Multi-region	two windows	D	Sdm	1	22	6.09	2.95	0.111	-62.4	weakly constrained
UGC05764 t	Multi-region	two windows	E	Im	2	10	3.79	1.81	0.027	-13.9	weakly constrained
ESO563-G021	Flattening window	0<p<2	C	Sbc	1	30	3.38	0.94	0.089	-63.3	-
IC4202	Flattening window	0<p<2	C	Sbc	1	32	1.31	0.57	0.051	-16.3	-
NGC0289	Unresolved (bulge / non-nested)	extreme window parameters	D	Sbc	2	28	2.70	0.73	0.080	-40.9	phenomenological
NGC2955	Unresolved (bulge / non-nested)	extreme window parameters	E	Sb	1	24	3.06	1.91	0.065	-20.2	phenomenological
UGC02487	Unresolved (bulge / non-nested)	extreme window parameters	D	S0	1	17	38.15	0.66	0.026	-553.7	phenomenological

Galaxy	Class		Sub-type	Cat.	Type	Q	n	χ^2_ν	S	χ^2_ν	χ^2_ν final	RMS final	Δ BIC	Quality / grade
UGC03546	Unresolved (bulge / non-nested)		extreme window parameters	D	Sa	1	30	3.40		0.80	0.058		-62.6	phenomenological
UGC06786	Unresolved (bulge / non-nested)		extreme window parameters	D	S0	1	45	1.41		0.31	0.018		-33.2	phenomenological
UGC09133	Unresolved (bulge / non-nested)		extreme window parameters	D	Sab	1	68	64.01		4.71	0.035		-3915.8	phenomenological
NGC3726 t	Unresolved (bulge / non-nested)		two-L non-nested	B	Sc	2	12	2.98		2.19	0.091		-7.3	flag N
NGC4013	Unresolved (bulge / non-nested)		two-L non-nested	B	Sb	2	36	5.52		2.36	0.118		-104.9	flag N
UGC02885	Unresolved (bulge / non-nested)		two-L non-nested	B	Sc	1	19	2.03		0.75	0.056		-17.3	flag N
UGC06614 t	Unresolved (bulge / non-nested)		two-L non-nested	B	Sa	1	13	1.62		0.33	0.054		-9.8	flag N
UGC06787	Unresolved (bulge / non-nested)		two-L non-nested	B	Sab	2	71	25.98		17.26	0.129		-627.4	flag N